



Modern MLOps

LECTURE

Defining MLOps

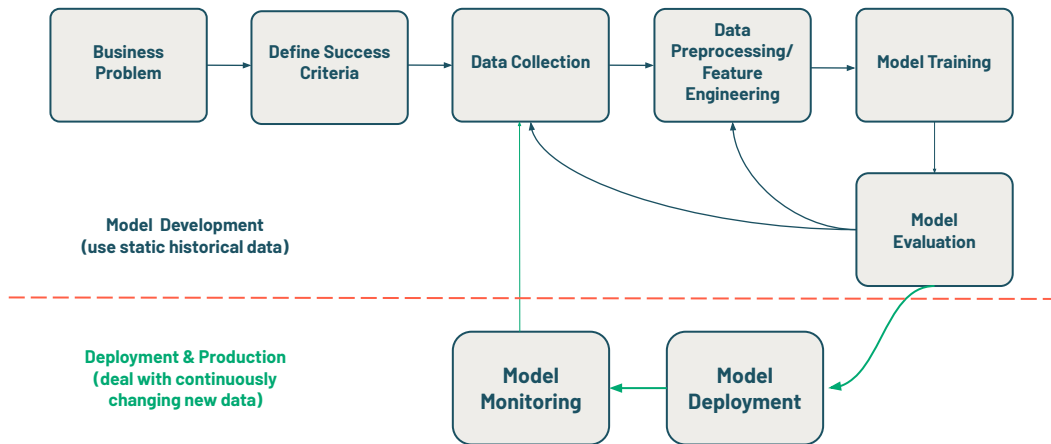


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In this lecture, we will discuss about Machine learning operations (**MLOps**), a rapidly **evolving field** where **building and maintaining** thorough, flexible and efficient workflows is critical.

The Machine Learning Full Lifecycle

End to End Process from Business Problem to Deployment and Monitoring

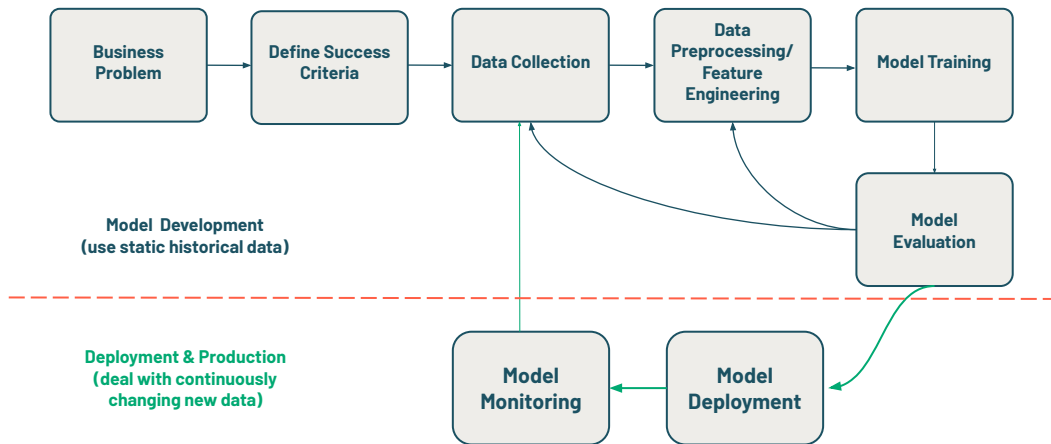


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- The **Machine Learning Lifecycle** = the end-to-end process from identifying a business problem to deploying and monitoring a model.
 - **Identify Business Problem** = Determine the **goal** for using machine learning.
 - **Define Success Criteria** = Set **quantifiable targets** to measure
 - **Data Collection and Preparation** = Gather, clean, preprocessing, feature engineering
 - **Model Training & Evaluation**
 - **Model Deployment**
 - Deploy **optimal models** to handle real-world operations and continuously changing data.
 - Set up infrastructure for robust deployment using Databricks.
 - **Model Monitoring**
 - Continuously track model performance.
 - Adjust and retrain model as necessary

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Continuous Improvement Cycle:

- **Feedback** from model performance **leads to** further **data collection**, **model retraining**, and **refinement**.

When working with machine learning, there are several steps you need to go through to transition from your initial business problem to a deployed model capable of making predictions.

The first step is defining your business problem. What is the goal you wish to achieve with machine learning? Once you have a clear understanding of the problem, you can proceed to define success criteria. Next, you will collect data to train your model. This data must be cleaned and preprocessed to ensure effective use. Once you have preprocessed data, you can move on to training the model. After training, it is crucial to evaluate its performance using a validation set. This helps you gauge how well the model is performing and identify any issues that need addressing.

Once you have a well-performing model, the next challenge is deploying it into production. Deploying a machine learning model can be challenging, as it involves managing the infrastructure for deployment and ensuring continued performance over time. Ongoing monitoring of the model's performance and making necessary adjustments are essential to maintain its effectiveness.

Defining MLOps

An all-inclusive, holistic approach to managing ML systems

The set of practices, processes, and technologies for managing **data**, **code**, and **models** to improve performance stability and long-term efficiency in ML systems.



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- **Overview:**
 - **MLOps is an all-inclusive** approach to managing ML systems.
 - Includes a comprehensive set of processes and automation techniques.
- **Core Components:**
 - **Data = Foundation** = requiring rigorous management to **ensure** accuracy, relevance, and security.
 - **Code = Backbone** = including algorithms and computational logic that need continuous integration and deployment.
 - **Models = Brain** = requiring lifecycle management from development through to deployment and monitoring.
- **Key Objectives:**
 - **Performance Stability:** Ensuring that ML systems perform consistently under varying conditions and data.
 - **Long-term Efficiency:** Optimizing resources and processes to sustain operations over time.

Understanding the Components of MLOps

MLOps components each address a specific part of ML projects

DataOps

A set of practices, processes, and technologies to organize and improve processes around data to increase speed, governance, quality, and collaboration.

DevOps

A set of practices, processes, and technologies to integrate and automate software development workflows.

ModelOps

A set of practices, processes, and technologies to organize and govern the lifecycle of machine learning and artificial intelligence models.



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- **DataOps:**
 - Focuses on the automation and monitoring of data pipelines.
 - **Benefits:** efficiency in data workflows, data quality and governance. collaboration.
- **DevOps:**
 - Tools and practices that facilitate **rapid, safe, and reliable model deployment and updates**.
 - Ensures machine learning code is managed with the same rigor as software development projects.
- **ModelOps:**
 - **lifecycle management of ML models** from development to deployment and maintenance.
 - ***Ensures models remain accurate and reliable throughout their operational life***
- **Real-World Example:** An online retail company implements MLOps to manage its recommendation engines by using DataOps for real-time customer data processing and personalized recommendations, DevOps for frequent algorithm updates aligned with market trends, and ModelOps for continuous model monitoring with automatic retraining when performance declines.

Responsibilities of MLOps Components

Key Functions of DataOps, DevOps, and ModelOps in MLOps

DataOps

- Optimized data **processing**
- Centralized data **discovery, management, and governance**
- Ensured data **quality**
- Traceable data **lineage and monitoring**

DevOps

- Machine learning is **code**
- Continuous integration and continuous deployment (**CI/CD**)
- **Version control** via Git
- Production-grade **workflows**
 - Orchestration
 - Automation

ModelOps

- Move beyond models as objects
- Treating model code as **software**
- Treating models as **data**
- Manage the **model lifecycle**



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- **DataOps Key Benefits:**
 - **Efficient Data Processing:** Utilizes powerful processing engines to handle *large* datasets effectively and optimized data storage and access.
 - *platforms like Apache Hadoop and Spark for managing and processing large volumes of data*
 - **Data Discoverability and Management:** Centralized data management allows consistent data governance Ensures easy access to relevant data.
 - **Data Quality Enforcement:** Implements **checks to validate data integrity** and accuracy
 - **Data Lineage and Monitoring:** Tracks data used in training models to meet compliance requirements.
 - Monitors to prevent degradation due to changing data trends.

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- **DevOps Key Concepts**
 - **Machine Learning as Code**= all aspects of ML, from **feature organization to model deployment**, are fundamentally code-driven processes.
 - **Machine learning projects are at their core about data and software**; DevOps addresses the software component
 - **Continuous Integration and Deployment (CI/CD)**:
 - incremental code updates, automated testing, and merging approved code into production environments.
 - *Jenkins or GitHub Actions: Popular tools for automating CI/CD pipelines.*
 - **Version Control**: Essential for tracking changes, managing updates, and ensuring that only verified code is deployed.
 - *Git: Used for version control to manage and track code changes.*
 - **Production-Grade ML Workflows**:
 - **Orchestration**: various components execute in the correct sequence.
 - **Automation**: operate efficiently in production without manual intervention.

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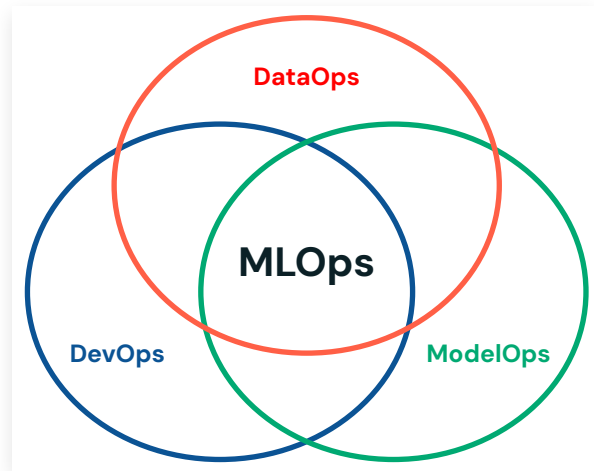
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- **ModelOps**
 - **Traditional Approach:** Treated models as static objects that move from development to testing and then to production.
 - **Vs. Modern Approach:** Views models as both code and data, integrating principles from both DataOps and DevOps.
 - **Key Aspects of Modern ModelOps:**
 - Models are dynamic, requiring continuous updates and management similar to how software is developed.
 - **Treating Model Code as Software:** version control, CI/CD practices
 - Applies data management techniques to model artifacts, ensuring data integrity, versioning, and compliance are maintained throughout the model's lifecycle.
 - **Managing the Model Lifecycle:**
 - Involves tracking development, validation, testing, and deployment phases to ensure models are effectively **transitioned through each stage.**

Comprehensive MLOps

Operationalizing the entire machine learning solution

The set of processes and automation for managing **data**, **code**, and **models** to improve performance stability and long-term efficiency in ML systems.



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- **Interconnection of Components:**
 - **The overlap of DataOps, DevOps, and ModelOps ensures that all aspects of ML systems are streamlined and efficient.**
 - **Data:** Models are only as good as the data they train on, requiring thorough data operations.
 - **Code:** Machine learning involves complex algorithms and computational workflows that must be managed like any software development project.
 - **Models:** encapsulate both code and data, requiring lifecycle management that spans across both DataOps and DevOps.
- **Benefits of Comprehensive MLOps:**
 - **Performance Stability:** Ensures that ML systems perform optimally under varying operational conditions.
 - **Long-term Efficiency:** Enhances the sustainability and cost-efficiency of ML systems over time.
 - **Scalability:** Facilitates the scaling of ML operations to meet growing data and operational demands.

A Simple Example ML Project

Retail Recommendation System

1. A business owner defines a problem to be solved with a recommendation service.
2. A data scientist begins **exploring governed data** associated with the service.
3. A data scientist **develops** a **scalable** ML solution using using relevant **data** while **tracking the experiment**.
4. A machine learning engineer implements **CI/CD, automates** the ML solution, and establishes **model performance monitoring**; the data is written to a **production catalog**.



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This project delivers a retail recommendation system to boost customer engagement and sales through personalized suggestions. The process begins with problem definition by the business owner, followed by data exploration where data scientists analyze governed data in Databricks using notebooks or AutoML. A scalable ML model is then developed with MLflow for experiment tracking to ensure performance and scalability. Next, a machine learning engineer implements CI/CD pipelines, automates deployment, and establishes performance monitoring. Finally, all operational data is cataloged in the production environment, enabling continuous model refreshes and retraining to maintain or improve effectiveness.

Why does MLOps matter?

Success depends on quality data and operations practices

- **Defining an effective strategy**

- ML systems built on quality data
- Streamlining the process of taking solutions to production
- Operationalizing performance and effectiveness monitoring

- **So what?**

- Time to realizing business value is accelerated
- Reduction in manual oversight by high-value data science teams

Real-world Example

Databricks customer CareSource accelerated their model's development and deployment, resulting in a **self-service MLOps solution for data scientists** that reduced ML project time from 8 weeks to 3–4 weeks.

The CareSource team can extend this approach to other machine learning projects, realizing this benefit broadly.

Learn more about the work here.



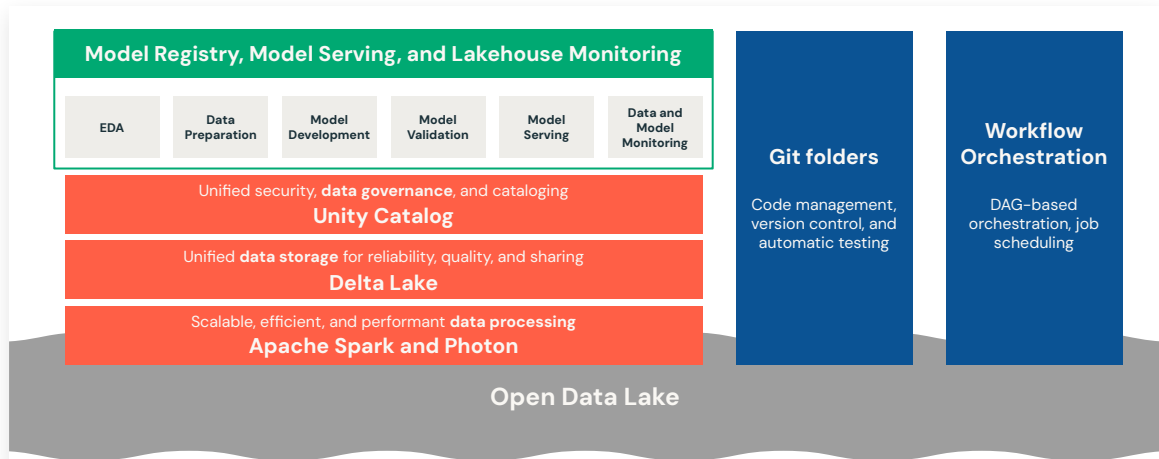
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- **Importance of a Defined Strategy:**
 - **Quality Data:** Fundamental to building effective ML systems.
 - **Streamlined Production Process:** Enhances the speed of taking solutions from development to deployment.
 - **Operational Performance and Effectiveness Monitoring:** Critical for maintaining system reliability and accuracy.
- **Business Impact:**
 - Faster transition from development to business impact.
 - Reduced Manual Oversight: Frees up valuable data science resources from routine maintenance to focus on innovation and strategic tasks.
- ***MLOps is not just a technical framework but a strategic advantage that can significantly enhance the efficiency and effectiveness of machine learning projects across sectors.***

A Single Platform for Modern MLOps

Combining **DataOps**, **DevOps**, and **ModelOps** solutions

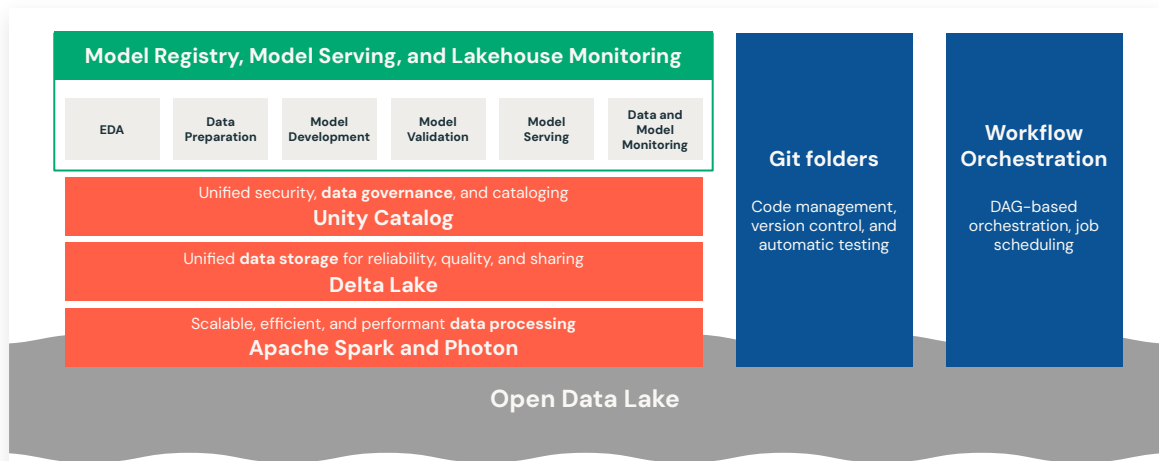


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DataOps leverages Apache Spark and Photon for scalable, efficient data processing, with Photon enhancing speed while remaining Spark-compatible. Unity Catalog ensures centralized governance, security, and organization of data assets across workspaces, while Delta Lake provides reliable storage that combines the strengths of data lakes and warehouses. On the DevOps side, Repos offer a Git-compatible environment for version control, branch management, and automated testing, supporting standard Git operations. Workflows use DAG-based orchestration and scheduling with tools like Jobs and Delta Live Tables to automate and streamline ML pipelines.

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ModelOps in Databricks unifies model storage, deployment, and monitoring through Model Registry, Model Serving, and Lakehouse Monitoring. For example, a telecommunications company uses these capabilities to manage churn prediction models—storing and versioning models in the registry, deploying them for real-time customer data streams, and monitoring performance to adapt to behavioral changes. Databricks provides a unified Data and AI platform that integrates DataOps, DevOps, and ModelOps, ensuring high-quality data, seamless automation, and efficient orchestration of ML workflows. This integration enables efficient pipeline management, reduces manual effort, and fosters collaboration and smooth role transitions across teams.



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